Round 1: overview

Platypus with one TEE call in place of the two zero-knowledge proofs
Serials are one-time public keys: no secret that can spend a user's funds reaches the enclave

Parties. red untrusted green runs attested code, may crash orange trusted for integrity, curious about privacy
Guarantees. bank rests on a check the bank does itself enclave key rests on the secrecy of the enclave's signing key
blue sealed to the attested enclave

Users (Alice pays, Bob receives)
May double spend, replay old states, forge states, collude with the bank

On the device
seed: never leaves the device; derives every one-time key and blind
state $S = (b,\ pk,\ r)$: balance, serial key, blind
bank signature σ on $C = \mathrm{Com}(b, pk, r)$: makes it money
one-time keys sk_i, pk_i: pk_i is the serial of the i-th state

To pay or receive
build the new state C' with the next key and blind
sign $m = H(pk_A, C'_A, pk_B, C'_B)$ with the spent state's sk
seal the opening of the current state to the enclave
Bob submits to the enclave, then settles with the bank
finish with Bob's receipt or the public log

Chain of ownership
C_0 holds pk_0 $\xrightarrow{\text{signed by } sk_0}$ C_1 holds pk_1 $\xrightarrow{\text{signed by } sk_1}$ C_2
Every next key sits in a state the bank signed only after checking the previous key's signature: only the seed's owner can spend the chain, and no identity is involved

sealed halves
signed result
Anonymous network hides the sender

Bank's machine: relays bytes, sees only ciphertext and timing

Enclave (attested code, run by the bank)
Runs the attested code correctly; may crash;
attested and registered by the bank at each boot;
keeps nothing between requests

Per boot, never stored
channel key: decrypts requests and sealed halves
signing key: signs results enclave key

Checks a payment
E1 both states carry a bank signature
E2 the same amount $v > 0$ on both halves
E3 $b_A - v \geq 0,\ \ b_B + v \leq B_{\max}$ enclave key
E4 two different serials
E5 new states open to $b_A - v$ and $b_B + v$ enclave key
E6 both owner signatures on m

Sees: the openings of both states
Never sees: a user's seed or signing key

settle, refresh, enroll, read the log (anonymous)

Bank: issues the money, runs the enclave, signs new states
Follows the protocol and signs only valid results; analyses all it sees; may collude with users; keeps its records

Records
registered enclaves (attested signing keys)
spent serials
the stored answer to every settled request
its own state, used to issue and redeem
Not in it: balances, amounts, identities

Checks a settlement
B1 result signed by a registered enclave enclave key
B2 recompute m; Alice's signature under pk_A, Bob's under pk_B bank
B3 both serials unspent; burn both together bank
B4 sign both new states; a retry gets the stored answer bank

Public transaction log (as in Platypus; anyone can read or mirror it)
per settlement: two spent serials, two new commitments, the owners' and the enclave's signatures, the bank's signatures
used to finish a payment without a receipt and to recover a wallet from its seed; holds nothing the bank does not already see

What the bank learns

Operation	Sees	Never sees
Payment	two spent serials, two new commitments, signatures, time	who, the amount, balances
Refresh	one spent serial, one new commitment, time	who, the balance
Enroll	one new commitment, time	who, the first serial

Who guarantees what, and what happens if the enclave's memory is read (its code still runs correctly)

Property	Enforced by	Rests on	If the enclave's memory is read
Ownership: nobody spends another user's state	owner signature checked by the bank (B2)	bank	**holds**: signing keys never leave the devices; the enclave only sees signed transactions
A payment cannot be redirected or changed	both owners sign the exact new states (m)	bank	**holds**: a different payee or amount changes m; a replay finds the serial spent
No double spend	spent serials (B3)	bank	**holds**
Nobody can burn another user's state	burning a serial needs its owner's signature	bank	**holds**
Balance: no money is created	conservation checked by the enclave (E3, E5), trusted through its signature (B1)	enclave key	**fails**: the attacker signs results that inflate *its own* states; it still cannot take anyone's money
Privacy toward the bank	the enclave opens states, the bank sees only serials and commitments	enclave	**fails** for the payments processed: amounts, balances, and a link to each participant's next payment

Integrity is not yet fully decoupled from the TEE: ownership is enforced by the bank, balance conservation still rests on the enclave's key.
Next step: let the bank check conservation without seeing values (zero-knowledge proofs as in Platypus, additive commitments with range proofs, or per-result attestation).